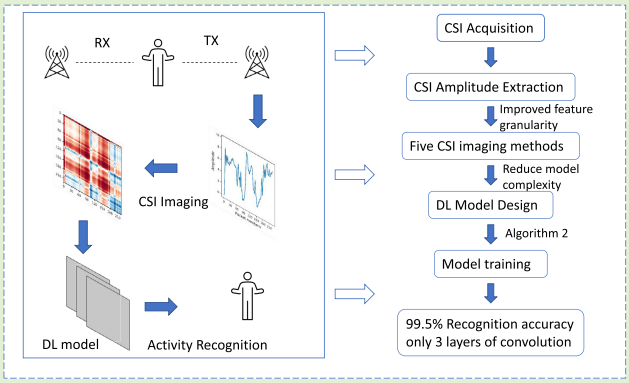
ImgFi: A High Accuracy and Lightweight Human Activity Recognition Framework Using CSI Image

Changsheng Zhang, *Graduate Student Member, IEEE*, and Wanguo Jiao

***Abstract*—Although human activity recognition (HAR) based on WiFi channel state information (CSI) has been widely studied, as the CSI signal is susceptible to the external environment, a recognition framework with high accuracy and low complexity remains a key and difficult problem. To improve WiFi sensing performance and reduce model complexity, ImgFi, a gesture recognition system, is proposed, which includes a conversion module and a recognition module. The conversion module is used to convert the CSI to images, and the key difference of the proposed ImgFi is using signal variations in arbitrarily different timestamps as feature extraction units rather than single signal points. This conversion can also take advantage of the powerful ability of the convolutional neural network (CNN) in the field of image recognition. Five CSI imaging approaches are introduced in ImgFi to realize the CSI conversion while a CNN is designed to recognize the activity from the converted images. To verify the proposed framework, we make some tests in our laboratory and get a dataset. The tests on four datasets under different contexts indicate that ImgFi with only three convolutional layers achieves 99.5% recognition accuracy. That is, our method has higher accuracy and lower complexity, which is important to related applications, such as smart homes and virtual reality. We will publish our code at https://github.com/shengjunzi/ImgFi.**



***Index Terms*—Channel state information (CSI), convolutional neural network (CNN), human activity recognition (HAR), image conversion, WiFi sensing.**

I. Introduction

HUMAN activity recognition (HAR) is a critical enabler for many applications, such as smart homes, healthcare, security surveillance, and virtual reality. The HAR is classified into three categories: sensor-based, vision-based, and wireless signal-based [1], [2], [3]. For the HAR based on sensors, the sensor must be mounted or worn on various parts of the body [4], which is less convenient than other categories. In the vision-based HAR system, some vision devices, such as cameras, need to collect video or image data [5], which has a privacy risk. With the rapid development of wireless communication, the HAR based on wireless signal has led to extensive research interest.

The wireless signal used by the HAR system includes radio frequency identification (RFID), ultra-wideband (UWB), and WiFi. However, the deployment of dedicated passive RFID tag arrays on the target for implementing RFID-based HAR approaches leads to increased deployment cost [6]. Meanwhile, the utilization of UWB for sensing is limited due to the lack of large-scale infrastructure deployment, whereas the reuse of the existing Wi-Fi devices at a much lower cost enables WiFi-based HAR approaches to be a convenient and easy-to-implement approach. Table I provides a summary and comparison of the characteristics of these HAR technologies. In particular, as the first international standard of integrated sensing and communication, IEEE 802.11bf works to set standards and promote the development of technology and application of sensing and recognition based on WiFi [7].

Two main types of information stored in WiFi signals are used for activity identification: the received signal strength indicator (RSSI) and the channel state information (CSI). The

Manuscript received 26 June 2023; accepted 14 July 2023. Date of publication 21 July 2023; date of current version 14 September 2023. This work was supported by the National Natural Science Foundation of China under Grant U22B2007 and Grant 61701241. The associate editor coordinating the review of this article and approving it for publication was Dr. Fabrizio Santi. *(Corresponding author: Wanguo Jiao.)*

The authors are with the College of Information Science and Technology, Nanjing Forestry University, Nanjing 210037, China (e-mail: sheng@njfu.edu.cn; wgjiao@njfu.edu.cn).

Digital Object Identifier 10.1109/JSEN.2023.3296445

1558-1748 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2026 at 12:51:29 UTC from IEEE Xplore. Restrictions apply.

TABLE I
OVERVIEW OF ACTIVITY RECOGNITION TECHNOLOGY

Technology	Convenience	Privacy	Cost	Accuracy
Sensor	Low	Moderate	High	Moderate-high
Cameras	High	Low	High	Moderate-High
RFRD	Low	Moderate	High	Moderate-high
UWB	Moderate	High	High	Moderate-high
WiFi	High	High	Low	Moderate-high

RSSI has limited sensing capability and low resolution, which makes fine-grained HAR difficult. The CSI can discriminate multipath characteristics and provide more detailed wireless channel information at the physical layer [8]. The existing WiFi-based HAR solutions primarily use the CSI [9], [10], [11], [12]. These works have proven that the HAR based on the CSI measured by WiFi devices is feasible.

In the CSI-HAR domain, early solutions relied heavily on traditional machine learning [13], [14], [15], [16]. In these solutions, the CSI signal is typically filtered by a Butterworth filter or wavelet transform to remove some of the noise. The statistical features are then manually extracted from the filtered signal to train the classifier. Such as WiChase [14], which extracts statistical data such as mean, standard deviation, 25th, and 75th percentiles, median absolute deviation, and maximum signal amplitude as input features for training classifiers. WiChase achieved a promising recognition accuracy of 94%. However, the reliance on traditional machine learning models on manual feature design and selection is a limiting factor in the application of WiFi-based HAR systems. This is because WiFi signal features are complex and variable over time, space, and environment, requiring extensive domain expertise and experience to design effective features. Moreover, the problem of WiFi sensing involves a large volume of data and complex nonlinear relationships, making accurate modeling with statistical or classical machine learning approaches challenging.

Due to the shortage of the classical machine learning in the WiFi-based HAR, some pioneers turned to various deep learning (DL) techniques, such as convolutional neural networks (CNNs) [17], [18], [19], [20], [21], [22], [23], recurrent neural networks (RNNs) [9], [24], [25], [26], and RNN + CNN [27], [28], [29], [30], to achieve end-to-end automatic extraction of signal features. The generalized framework of the existing DL-based CSI-HAR solutions is shown in Fig. 1. The CSI-HAR is regarded as a classification task, and researchers tend to design complex DL structures to obtain a high accuracy rate. However, due to the complexity of the mapping relationship between CSI signal changes and different human behaviors, as well as the different application domains of various DL models, the existing DL-based solutions, while effective in recognizing human activities from CSI, have some drawbacks that prevent them from improving recognition performance further.

Specifically, the advantage of 2-D CNN (2D-CNN) lies in its ability to process image features. When CNN is used, the CSI signal is treated as a simple image feature, with the length and width of the image representing spatial locations. However, the two dimensions of the CSI, namely, the timestamp and the channel state value, possess different semantics. Thus, treating

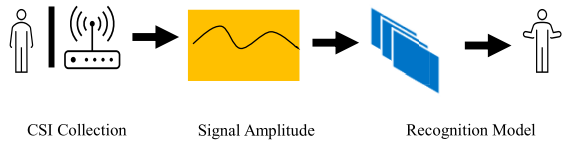


Fig. 1. Generalized DL framework for CSI-HAR.

CSI data as image input to the CNN network results in the loss of time series (TS) features. To achieve high recognition accuracy, a more complex CNN network must be constructed, requiring larger parameters. For example, Li et al. [20] created a two-stream convolution with high complexity, achieving 98.7% accuracy.

In contrast, RNN is ideal for handling TS features in CSI-HAR. However, CSI has temporal and spatial features, describing the combined effects of various factors on the signal propagation path. A single RNN framework cannot adequately model CSI at a fine-grained level, and the recognition accuracy of using LSTM alone is merely 90.5% [9]. Thus, combining RNN with other methods is necessary to achieve high recognition accuracy. A recent study by Schäfer et al. [25] successfully combined LSTM with support vector machine (SVM), resulting in an accuracy of 97%.

Furthermore, some works have attempted to combine CNN and RNN to achieve spatiotemporal modeling of the CSI. For instance, Yang et al. [30] used a combination of convolutional and bidirectional RNNs for gesture recognition and achieved 92.8% recognition accuracy. In contrast, Wi-Sensing proposed in [27] used a short-time Fourier transform (STFT) to convert CSI data into image data, achieving 0.96 accuracies by combining CNN and LSTM. However, the accuracy of these solutions relies on complex feature processing and learning models. It is worth noting that the majority of the existing DL-based CSI-HAR solutions extract features using a single time-stamped channel state value as the basic unit, which lacks information granularity. HAR based on WiFi signals involves establishing a mapping relationship between these signal variations and different human behaviors. More meaningful features are typically expressed as signal variations between timestamps rather than signal values for a single timestamp.

In this article, we propose a new CSI-HAR framework, ImgFi, to solve the above problem. Taking advantage of the breakthrough of CNNs in image recognition [31], [32], [33], [34], [35], we try to formulate the CSI-HAR problem as an image recognition problem. Thus, in the proposed ImgFi, a conversion module is added before the CNN. Using this module, the original CSI is converted into an image while the TS property of the CSI is preserved in the image. Four novel methods are introduced by ImgFi to define the amplitude fluctuations between arbitrary timestamps and realize the transformation from the CSI to the image. Besides, the traditional STFT method can also be used as the conversion method in ImgFi. Then, a 2-D CNN (2D-CNN), which has powerful image feature extraction capability of modeling CSI in space-time, is designed to find out the activity information from the converted image. The contributions made in this article are concluded as follows.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2026 at 12:51:29 UTC from IEEE Xplore. Restrictions apply.

1) We propose a standardized framework of the CSI-HAR, which meets both high accuracy and low model complexity requirements.
2) Five methods, recurrence plot (RP) transformation, Gramian angular sum field (GASF) transformation, Gramian angular difference field (GADF) transformation, Markov transition field (MTF) transformation, and STFT, are first introduced into the field of CSI imaging. The comparative analysis on these five methods is provided to demonstrate the advantage and shortage of the five imaging methods in processing CSI imaging.
3) Based on the property of the CSI images and the analysis of extraction operation, a 2D-CNN with three convolutional layers is designed. The proposed 2D-CNN can achieve comparable performance to mainstream architectures, e.g., ResNet50 and VGG19, while requires fewer parameters and floating point operations (FLOPs).
4) Some experiments are added, which further enrich the CSI datasets. The evaluation on four datasets with different environments including three public datasets and our added dataset demonstrates that the accuracy of the proposed ImgFi reaches 99.5% on four datasets while it has lower complexity, which indicates that the proposed ImgFi can be generally used.

The remainder of this article is organized as follows. We discuss related work in Section II. Section III provides an overview of the proposed ImgFi. Section IV specifies and explores the conversion method used for CSI imaging while Section V describes the proposed CNN architecture. Section VI describes the relevant experimental details. The analysis of the evaluation results is given in Section VII. Finally, Section VIII concludes the whole article.

II. Related Work

Our framework includes image transformation techniques and the DL model. To clarify our work, we first introduce the common image transformation method for 1-D TS and the existing CSI-HAR DL frameworks before presenting the proposed framework.

A. Image Conversion for 1-D TS

A 1-D TS refers to a sequence of data values arranged in chronological order, which can be represented mathematically as follows:

$$\mathbf{TS}_{1\times T} = (x_1, x_2, \ldots, x_T) \tag{1}$$

where x_t represents the data value at time point t, and T represents the length of the sequence. In most cases, x_t can be a real or complex number. Unlike other types of data, TS contains the temporal correlation which must be considered when classification algorithms work.

Wang and Oates [34] proposed three methods for converting TS data into images, GADF, GASF, and MTF. All three methods were recently used as an indirect method for TS data classification using CNNs. In [36], the Gramian angular fields (GAFs) are first used in activity recognition. Eckmann et al. [37] first applied the RP to the TS-to-image conversion. Ahmad and Khan [38] demonstrated the benefits of 2-D CNNs for image feature extraction to recognize the activity after using the RP to convert the inertial sensor data into images. However, there is no work using these methods to realize the image conversion for the CSI, especially for the CSI-HAR. Only the STFT has been used to do the conversion of the CSI in [27]. When introducing these methods into the system shown in Fig. 1, the CSI–HAR problem becomes interesting.

B. DL Framework for the CSI-HAR

There are three major DL frameworks for the CSI-HAR, RNN [9], [24], [25], [26], CNN [17], [18], [19], [20], [21], [22], [23], and RNN + CNN [27], [28], [29], [30]. The RNN is a class of RNNs that take the sequence data as the input and recurve in the evolutionary direction of the sequence, and all the nodes are connected in a chain-like manner. Among them, bidirectional RNNs (Bi-RNNs) and LSTM are two common RNNs. When using RNN, the CSI is usually considered a TS feature. The CNN has unique advantages in speech recognition and image processing because of its special structure of local weight sharing, especially images with multidimensional input vectors that can be fed directly into the network. This structure makes the CNN avoid the complexity of data reconstruction during feature extraction and classification. When the CNN is used, the CSI is considered as an image, and the relationship between the temporal and amplitude is treated as the spatial feature of the image. In the field of CSI-HAR, the spatiotemporal nature of the CSI has motivated the exploration of combined RNN and CNN models. Typically, the CNN is leveraged to extract spatial features from the CSI, which are then further refined using the RNN to extract temporal features. However, the combined approach increases the model complexity and training cost.

III. Overview of the ImgFi

To efficiently and precisely recognize the human activity from the CSI measured by the off-the-shelf WiFi devices, we propose a lightweight activity recognition system with high recognition accuracy, which is named as ImgFi. As illustrated in Fig. 2, taking the received signals affected by the user in the monitoring area as the input, ImgFi uses the CSI imaging module to process the input and then inputs the processing results to the activity recognition module. By comparing Fig. 1 with Fig. 2, ImgFi adds a CSI imaging module before the activity recognition module, which is used to preprocess the CSI before it is used to extract the activity information.

As shown in Fig. 2, on a wireless channel, the relationship between the received signal $\mathbf{Y}$ and the transmitted signal $\mathbf{X}$ is expressed as follows:

$$\mathbf{Y} = \mathbf{HX} + \mathbf{N} \tag{2}$$

where $\mathbf{H}$ is the channel gain matrix, and $\mathbf{N}$ is the noise matrix. The CSI is the value of each element in the matrix $\mathbf{H}$. Each element is represented in the complex form $\alpha_t e^{-j\theta_t}$ where α_t and θ_t correspond to the magnitude and the phase at moment t, respectively.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2026 at 12:51:29 UTC from IEEE Xplore. Restrictions apply.

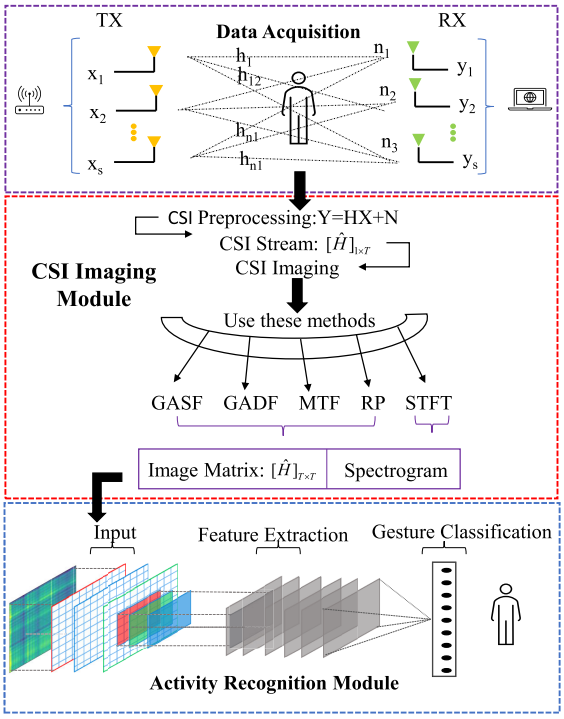


Fig. 2. Framework of the proposed ImgFi.

Given a system of transceiver antennas, we refer to the CSI value over time at a given carrier frequency as the CSI stream, which is denoted as $\mathbf{H}_{1\times T}$. As the activity information lies in the magnitude variation, the amplitude of the CSI stream should be extracted. The extracted CSI stream is denoted as $[\hat{\mathbf{H}}]_{1\times T}$, which is the input of the imaging module.

The CSI imaging module converts the 1-D amplitude variation sequences into an image by computing the amplitude relationship between any two-time stamps. However, the framework using different conversion methods achieves different performances. As shown in Fig. 2, five methods are introduced in ImgFi to realize the imaging separately.

The HAR module implements a CNN for activity recognition. The fundamental CNN is made up of three structures: convolution, activation, and pooling. The CNN output is a unique feature space for each image. Then this feature space is inputted into the fully connected layer undertaking the image classification task, which completes the mapping from the input image to the label set. Taking the RP-transformed image as an example, the spatiotemporal features of the CSI are extracted by a feature extractor, and then the features are classified in a classifier. The extracted features are used to determine the user's activity type using a fully connected layer. To clarify the proposed framework, we will introduce the conversion module in Section IV and the recognition module in Section V in detail separately.

IV. CSI Imaging Module

The main function of the CSI imaging module is converting the CSI data into images which can be used to train CNNs. To realize this function and demonstrate the advantage of this conversion, we provide five different methods here.

A. GAF Transformation

The GAF method comprises two techniques, GASF and GADF, both of which calculate the point of the image using the polar plot. According to the definition of the Gram matrix, the CSI value should be scaled to the 0–1 interval. Thus, the min–max normalization is used to normalize each element in the matrix $[\hat{\mathbf{H}}]_{1\times T}$ before converting it into the polar form. The corresponding angle is calculated as follows:

$$\theta_i = \arccos(\hat{H}_i), 0 \le \hat{H}_i \le 1. \tag{3}$$

When the GASF is adopted, the transformed image is expressed by a matrix which is denoted by **GS** given in (4). The matrix **GS** consists of the cosine function of the sum of any two angles with size $T \times T$

$$\mathbf{GS} = \begin{bmatrix} \cos(\theta_1+\theta_1) & \cdots & \cos(\theta_1+\theta_T) \\ \cos(\theta_2+\theta_1) & \cdots & \cos(\theta_2+\theta_T) \\ \vdots & \ddots & \vdots \\ \cos(\theta_T+\theta_1) & \cdots & \cos(\theta_T+\theta_T) \end{bmatrix}. \tag{4}$$

Similarly, the image of the GADF transformation is denoted by the matrix **GD** shown in (5) which consists of the sine function of the difference of any two angles. The matrix **GD** has the same size as the matrix **GS**

$$\mathbf{GD} = \begin{bmatrix} \sin(\theta_1-\theta_1) & \cdots & \sin(\theta_1-\theta_T) \\ \sin(\theta_2-\theta_1) & \cdots & \sin(\theta_2-\theta_T) \\ \vdots & \ddots & \vdots \\ \sin(\theta_T-\theta_1) & \cdots & \sin(\theta_T-\theta_T) \end{bmatrix}. \tag{5}$$

B. Short-Time Fourier Transformation

The STFT is a general tool for signal processing. To process the original signal $\hat{\mathbf{H}}$, a window function $W(t)$ is first defined. The window function used in this article is the Hanning window, which is expressed as follows:

$$W(t) = \frac{1}{2}\left[1 - \cos\left(\frac{2\pi\ (t-1)}{N}\right)\right] \tag{6}$$

where N denotes the length of the window.

Then, move the window to the starting point of the signal. When the center point of the window function is $t = \tau_0$, the signal is windowed and is expressed as follows:

$$y(t) = \hat{H}_t W(t-\tau_0). \tag{7}$$

And then, the Fourier transform is performed

$$S(\omega, \tau_0) = \sum_{t=0}^{N} \hat{H}_t W(t-\tau_0) e^{-j\omega t}. \tag{8}$$

This yields the spectral distribution of the first segmented sequence $S(\omega, \tau_0)$. It represents the spectral result of the transformation of the original signal when the center of the window function is τ_0. Move the center of the window function to τ_1; the moving distance is generally smaller than the width of the window. Thus, there is a certain overlap between the front and rear windows.

Repeat the above operation, keep sliding the window, and finally get the spectrum results of all the segments from τ_0 to τ_m. We set the window length N to be 10 and the overlap

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2026 at 12:51:29 UTC from IEEE Xplore. Restrictions apply.

length of the window to be 5. The size of each segmented sequence is $N/2+1$, and the total sliding window is m times. Finally, we get the spectrum matrix expressed as follows:

$$S = (S_{\tau_0}, S_{\tau_0}, \ldots, S_{\tau_m}) \tag{9}$$

where m equals $T-5/N-5$.

C. MTF Transformation

The MTF is a technique for converting 1-D TS data into 2-D images, and the features extracted by this method can represent dynamic changes in time and frequency. The MTF divides the magnitude data $[\hat{H}]_{1\times T}$ into Q states according to the range of values taken, and each data point $\hat{H}_t$ belongs to a unique q_i. We can construct a Markov transfer matrix W of size $Q \times Q$. W_{ij} denotes the transfer probability from state q_i to q_j. W_{ij} is given by the count of states q_i following q_j divided by the number of states q_i. The formula is expressed as follows:

$$W_{ij} = P(q_i|q_j) = \frac{P(q_i)}{P(q_j)}. \tag{10}$$

The Markov transfer matrix is less sensitive to the distribution of the original data and loses temporal information. To deal with this loss, we further expand the transfer probabilities associated with the temporal locations and convert the Markov transfer matrix into an MTF which is expressed as follows:

$$\mathbf{M} = \begin{bmatrix} W_{ij|\hat{H}_1\in q_i,\hat{H}_1\in q_j} & \cdots & W_{ij|\hat{H}_1\in q_i,\hat{H}_t\in q_j} \\ W_{ij|\hat{H}_2\in q_i,\hat{H}_1\in q_j} & \cdots & W_{ij|\hat{H}_2\in q_i,\hat{H}_t\in q_j} \\ \vdots & \ddots & \vdots \\ W_{ij|\hat{H}_t\in q_i,\hat{H}_1\in q_j} & \cdots & W_{ij|\hat{H}_t\in q_i,\hat{H}_t\in q_j} \end{bmatrix}. \tag{11}$$

The size of $\mathbf{M}$ is $T \times T$, and M_{ij} is the probability that the state in which $\hat{H}_i$ is located is transferred to the state in which $\hat{H}_j$ is located.

D. RP Transformation

An RP is a nonlinear data analysis technique which reveals the internal structure of a TS and provides the prior knowledge of similarity and information content. The RP transformation transforms a 1-D TS into a visual matrix which can be used to check the smoothness and intrinsic similarity of the TS. Take the channel amplitude data $[\hat{\mathbf{H}}]_{1\times T}$ as the input and use (12) to calculate the absolute value of the difference between two points, which represents the distance between two points

$$\mathbf{H}_R = \begin{bmatrix} ||\hat{H}_1 - \hat{H}_1|| & \cdots & ||\hat{H}_1 - \hat{H}_t|| \\ ||\hat{H}_2 - \hat{H}_1|| & \cdots & ||\hat{H}_2 - \hat{H}_t|| \\ \vdots & \ddots & \vdots \\ ||\hat{H}_t - \hat{H}_1|| & \cdots & ||\hat{H}_t - \hat{H}_t|| \end{bmatrix}. \tag{12}$$

Subtract the predefined threshold ε, the distance is converted into a binary number using the Heaviside function Θ. The formula is expressed as follows:

$$\mathbf{RP} = \Theta(\mathbf{H}_R - \varepsilon). \tag{13}$$

The converted result **RP** is an image of the size $T \times T$. If the channel amplitude cannot be reduced to two states, the threshold and the Heaviside function are replaced by making the matrix nonbinarized. For the value of threshold ε, it is recommended to set ε to 0. Then, more information of the original data's internal structure is preserved. For RT transformation, the amplitude relationship between any two timestamps is expressed as the absolute value of the difference between two points.

E. Discussion of the Conversion

Images are created using the imaging algorithm to represent a series of CSI data. Figs. 3 and 4 show the raw CSI and its transformed images, which correspond to the falling and clapping actions, respectively.

For the GASF, according to (3), the ith row jth column element of matrix **GS** is calculated as follows:

$$\cos(\theta_i + \theta_j) = \hat{H}_i\hat{H}_j - \sqrt{1-\hat{H}_i^2}\sqrt{1-\hat{H}_j^2}. \tag{14}$$

From (14), it can be found that the larger the values of $\hat{H}_i$ and $\hat{H}_j$, the larger the value of $\mathbf{GS}_{(i,j)}$ is. As shown in Figs. 3(b) and 4(b), the amplitude relationship between any two timestamps is reflected by colors. For example, in Fig. 3(a), the amplitudes at 90 and 150 on the horizontal axis are at the lowest global position, whereas in Fig. 3(b), the response is reddish (90, 150). The magnitudes at 90 and 150 on the horizontal axis are at the higher global position in Fig. 4(a), and the color at (90, 150) is bluish in Fig. 4(b).

We can see the benefit of this transformation by visualizing it. When the CNN model is used to extract features from GASF images, the amplitude change relationship with different timestamps is the unit of the generated features at this time, which provides richer and more meaningful information. Furthermore, the GASF image retains the original CSI's time dependence from top left to bottom right.

Similarly, according to (15), the larger the difference between $\hat{H}_i$ and $\hat{H}_j$, the larger positive or negative difference of the GADF image

$$\sin(\theta_i - \theta_j) = \hat{H}_j\sqrt{1-\hat{H}_i^2} - \hat{H}_i\sqrt{1-\hat{H}_j^2}. \tag{15}$$

Compared with GASF, GADF is more sensitive to the volatility of signal changes, and the regions with distinct colors are where the raw CSI values fluctuate more. For example, in the raw CSI, the amplitudes located at 90 and 110 have large fluctuations. Fig. 3(c) shows a bright blue color at (90, 110). The region with distinct color in Fig. 4(c) is much smaller than Fig. 3(c), which can be explained by the original CSI. As shown in Figs. 3(a) and 4(a), the applause produces signal fluctuations that are more subdued than falls.

STFT, unlike the other four, can perform time–frequency analysis on CSI signals. As shown in Figs. 3(d) and 4(d), STFT can reflect the amplitude (the larger the amplitude, the bluer the color) of the CSI in the frequency domain over time. The created MTF images are shown in Figs. 3(e) and 4(e). With increasing blueness in the MTF, the possibility of a shift between two points increases. In the original CSI, for example,

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2026 at 12:51:29 UTC from IEEE Xplore. Restrictions apply.

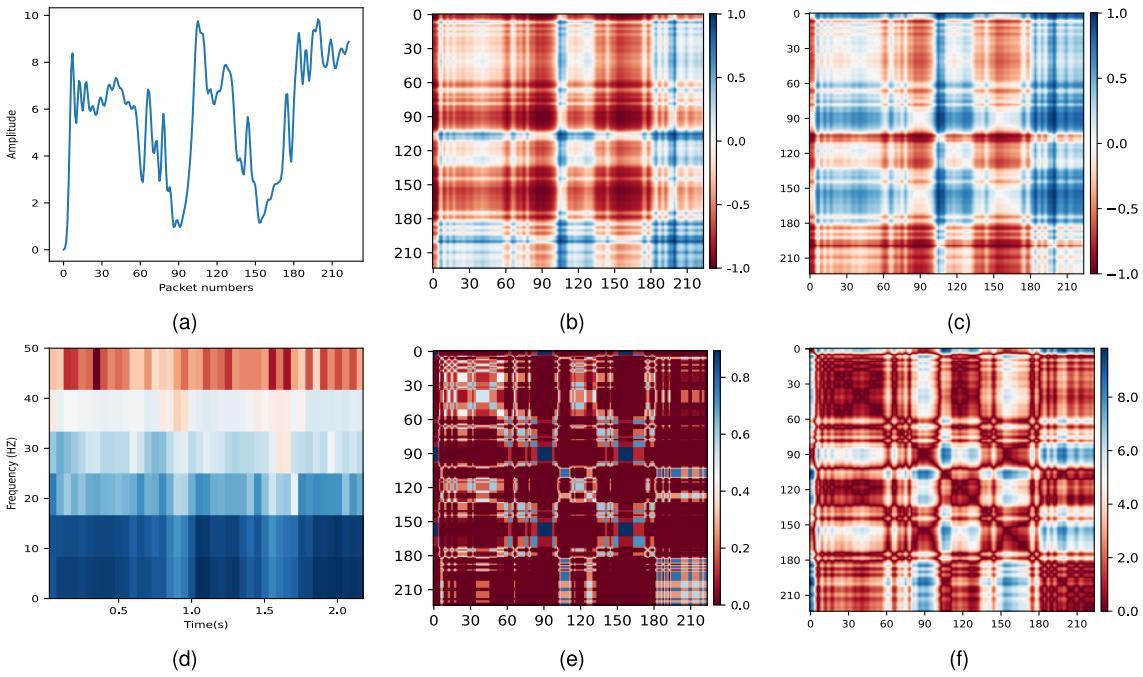


Fig. 3. Single subcarrier CSI and conversion images corresponding to falling action. (a) CSI. (b) GASF. (c) GADF. (d) STFT. (e) MTF. (f) RT.

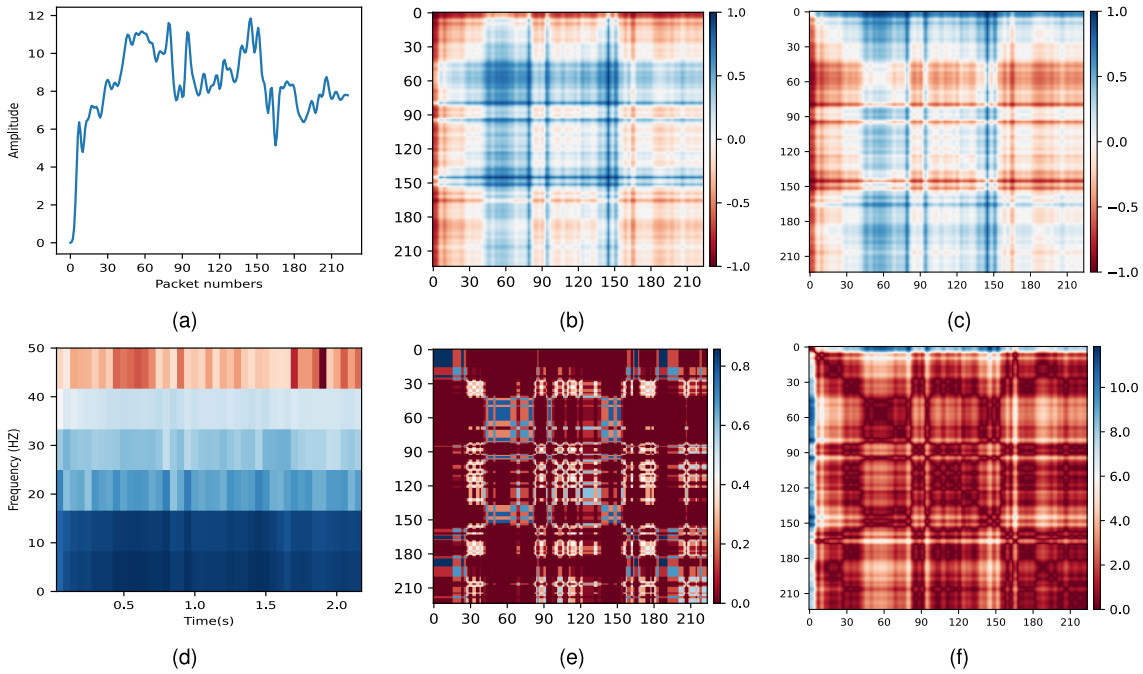


Fig. 4. Single subcarrier CSI and conversion images corresponding to clapping action. (a) CSI. (b) GASF. (c) GADF. (d) STFT. (e) MTF. (f) RT.

the two points at (90 and 50) are at the lowest position of the magnitude value and have a high similarity, resulting in a high degree of blue in the MTF image. The transformed RT images are shown in Figs. 3(f) and 4(f). The RT is similar to GADF which is more sensitive to signal volatility. The GAF transformed image is calculated using the polar angle, whereas the MTF is calculated using the Markov chain, which may lose some information. The RT is directly transformed from the original CSI value, resulting in less information loss.

From the above discussion, it can be found that the imaging conversion has the following advantages.

1) The basic unit of feature extraction changes from a single amplitude value to a two timestamp amplitude change value, and it has a richer information granularity.
2) The temporal dependence of CSI is still maintained from the top left corner to the bottom right corner of the image.
3) The imaging is adapted to the feature extraction advantage of 2D-CNN for images, in addition to providing more information.

Through comparing these five imaging methods, there are some differences.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2026 at 12:51:29 UTC from IEEE Xplore. Restrictions apply.

1) Compared with the RP and STFT transforms, the GAF transformed images are calculated based on polar angles, while the MTF is calculated based on Markov chains, which may lose some information.
2) For the MTF conversion, the state number Q can be changed as a hyperparameter and has an impact on the result. The window length and overlap length are also adjustable parameters for STFT. In contrast, RT and GAF conversions are relatively easy.
3) Compared with other four methods, the STFT is a traditional signal imaging method used to analyze the time–frequency property of CSI signals.

V. Activity Recognition Module

A lightweight CNN model is designed in ImgFi to mine the spatial–temporal features of CSI images. The overall structure of the proposed model is depicted in Fig. 4. The model is composed of a feature extractor with four convolutional layers and a classifier with fully connected layers. There are batch normalization, activation, and pooling layers between every two convolutional layers. Instead of the traditional flattening operation, the four convolutional layers are followed by an adaptive average pooling layer. Finally, the fully connected layers, in conjunction with the dropout and activation layers, serve as classifiers, integrating the learned features, classifying them, and computing the loss values using the CrossEntropyLoss function.

A. Feature Extraction

Convolutional layers are the core building blocks of the CNN, and we use four convolutional layers to process the input feature images of the network. As shown in Fig. 5, the size of the convolution kernel is 3×3 and the number of channels is 64, 128, 256, and 512. The stride of the second, third, and fourth convolutional layers is set to 2 for downsampling to reduce computation. When the feature image passes through the convolutional layers, the output is represented as follows:

$$Z_c^{(l)} = W_c^{(l)} * A_c^{(l-1)} + b_c^{(l)} \tag{16}$$

where $A_c^{(l-1)}$, $W_c^{(l)}$, $Z_c^{(l)}$ denote the input, weight matrix, and output of the l-layer channel c, respectively, $b_c^{(l)}$ denotes the bias vector, and $*$ denotes the convolution operation.

After the convolutional layer, we add a BN layer to improve the convergence speed of the network [39]. This layer first normalizes the input $Z^{(l)}$ by calculating the mean μ_c and variance σ_c^2 of a batch, which are shown as follows:

$$\mu_c = \frac{1}{B}\sum_{i=1}^{B} Z_c^{(l)(i)} \tag{17}$$

$$\sigma_c^2 = \frac{1}{B}\sum_{i=1}^{B}\left(Z_c^{(l)(i)} - \mu_c\right)^2 \tag{18}$$

where B is the batch size, which indicates the number of input images. We introduce the scaling parameter δ and the translation parameter γ to ensure that the original learned features are kept after each data normalization. Then, the output is expressed as follows:

$$\tilde{Z}_c^{(l)} = \gamma\frac{Z_c^{(l)} - \mu_c}{\sqrt{\sigma_c^2 + \epsilon_1}} + \delta \tag{19}$$

where δ and γ are learnable parameters which are changed during network training, and ϵ_1 is a constant which is used to prevent improper computation due to zero variance. During normalization, the bias term b can be ignored because the normalization operation subtracts the mean value.

We use the rectified linear unit (ReLU) function as an activation layer to threshold each input element. ReLU sets any value less than zero to zero and leaves any value greater than zero unchanged. It reduces the interdependence of parameters and increases the nonlinear relationship between the layers of the neural network. After the activation layer, the feature image is represented as follows:

$$A_c^{(l)} = r_c^{(l)}\tilde{Z}_c^{(l)} \tag{20}$$

where r denotes the ReLU function.

Finally, to further reduce the computation, we use the maximum pooling layer to extract the maximum value of the rectangular region around the tensor. The trait of pooling can effectively eliminate unnecessary information. We give the complete image feature extraction process for a particular convolutional layer in Algorithm 1.

Algorithm 1 Feature Extraction

INPUT: $l-1$ layer channel c feature image $A_c^{(l-1)}$
OUTPUT: l layer channel c feature image $A_c^{(l)}$
Initialize: Batch size $B = 128$, Feature image size $= H \times W$

Convolution operation: $Z_c^{(l)} = W_c^{(l)} * A_c^{(l-1)} + b_c^{(l)}$
Batch normalization:
Calculate the mean value: $\mu_c = \frac{1}{B}\sum_{i=1}^{B} Z_c^{(l)(i)}$
Calculate the variance: $\sigma_c^2 = \frac{1}{B}\sum_{i=1}^{B}(Z_c^{(l)(i)} - \mu_c)^2$
Linear transformation: $\tilde{Z}_c^{(l)} = \gamma\frac{Z_c^{(l)} - \mu_c}{\sqrt{\sigma_c^2+\epsilon}} + \delta$
Activation function: $A_c^{(l)} = r_c^{(l)}\tilde{Z}_c^{(l)}$
Pooling: $A_{c,i,j}^{(l)} = max(A_{c,i,j}^{(l)}, A_{c,i+1,j}^{(l)}, A_{c,i,j+1}^{(l)}, A_{c,i+1,j+1}^{(l)})$
$i \le (H-2)/2,\ j \le (W-2)/2$
RETURN: $A_c^{(l)}$

B. Feature Classification

With the kernel size $k \times k$ of the pooling layer, stride s, padding p (default 0), and the input image size $\text{input}_{\text{size}}$, the output size is computed as follows:

$$\text{output_size} = (\text{input_size} - k + 2 * p)/s + 1. \tag{21}$$

In turn, the output size can be used to calculate the core and stride. The adaptive averaging pooling layer's core size and stride can be adjusted to accommodate the given output size. The output of the convolution layer must be flattened before it can be sent to the fully connected layer. After four layers of

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2026 at 12:51:29 UTC from IEEE Xplore. Restrictions apply.

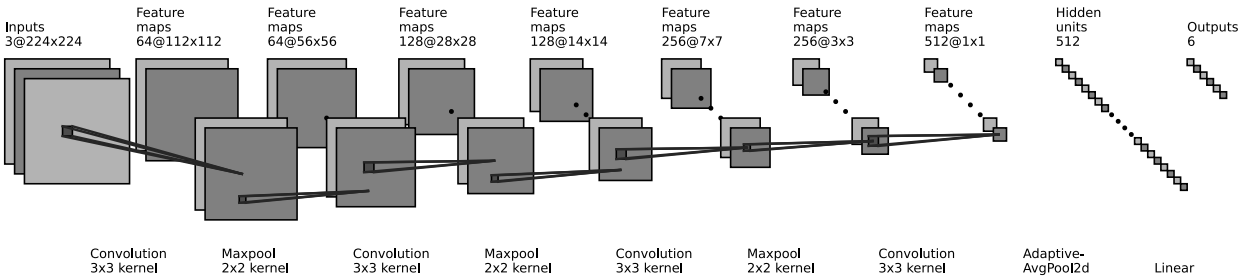


Fig. 5. Specific structure of the proposed CNN.

convolution, we use an adaptive average pooling layer instead of the standard flattening operation to designate the output size as 1×1, which can reduce computing cost. The formula for average pooling is expressed as follows:

$$A_{ij}^{(l)} = \frac{1}{|R_{ij}|} \sum_{(p,q)\in R_{ij}} x_{pq} \tag{22}$$

where $A_{ij}^{(l)}$ denotes the average pooled output value of the feature map in the rectangular region R_{ij}, x_{pq} denotes the elements located at (p, q) in the rectangular region R_{ij}, and $|R_{ij}|$ denotes the number of elements in the rectangular region. As shown in Fig. 5, the input dimension of the fully connected layer at this point is 512. To avoid overfitting, we use the dropout layer to eliminate features obtained by feature extraction from the network with a particular probability during the network training process. Finally, after passing through the activation layer we use the fully connected layer to linearly combine the input data to determine the likelihood of the expected output class. The final output of the fully connected layer is $(x_1, x_2, \ldots x_n)$, where n is the number of classifications.

C. Model Training

During DL backpropagation, the optimizer directs the loss function parameters to update the appropriate size in the right direction, such that the updated parameters keep the loss function value close to the global minimum. To compute the loss values, we use the CrossEntropyLoss function. First, we convert the output of the fully connected layer into a probability distribution in the range [0,1] with a sum of 1. The calculation is done by taking the exponent of the natural constant e for each value x_i of the input vector and dividing it by the sum of the exponents of the base e corresponding to each value x_j. Then logarithms are used to solve the function overflow and underflow problems. Finally, we invert the mean by multiplying the true label y_i of the data by the activation result after encoding it with one_hot. The formula is

$$L(y_i, x_i) = -\frac{1}{B} \sum_{i=1}^{B} y_i \log \frac{e^{x_i}}{\sum_{j=1}^{n} e^{x_j}}. \tag{23}$$

We use Adam's algorithm as an optimizer to update the model parameters. First, we calculate the gradient g_t of the loss function L with respect to the current parameters W_t, expressed by the formula

$$g_t = \nabla L(W_t) \tag{24}$$

TABLE II
EXPERIMENTAL PARAMETER SETTINGS

Parameter	parameter setting
Dataset partitioning	Training/validation/test(0.7/0.15/0.15)
Loss function	CrossEntropyLoss
Optimizer	Adam(β_1 = 0.9,β_2 = 0.99,ϵ_2 = 1e-9)
Learning rate	λ = 0.001,decay rate = 0.8
Batch size	B = 128
Epoch	20

where t represents the current number of iterations. The first-order exponential moving weighted average m_t and the second-order exponential moving weighted average v_t of the historical gradients are then calculated. The moving average decay rate is controlled by the hyperparameters β_1 and β_2

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \tag{25}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2. \tag{26}$$

The deviation correction for m_t and v_t is expressed as follows:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \hat{v}_t = \frac{v_t}{1 - \beta_2^t}. \tag{27}$$

The final update parameter is expressed as follows:

$$W_{t+1} = W_t - \frac{\lambda \hat{m}_t}{\sqrt{\hat{v}_t + \epsilon_2}} \tag{28}$$

where λ is the learning rate, and ϵ_2 is the smoothing term to prevent the denominator from being zero. In Table II, we can obtain the specific parameter settings.

In Algorithm 2, we give the complete process of learning the model parameters.

VI. EXPERIMENTAL DETAILS

To develop and evaluate our model, we give specific details on data acquisition and evaluation.

A. Data Details

To verify the generalization of ImgFi, we used three public datasets, that is, Wiar, SAR, and Widar3.0, and our self-test dataset. This section describes the experimental environment for collecting CSI and the background of three public datasets.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2026 at 12:51:29 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2 Model Training Methods

INPUT: Feature image :P = $(p_1, p_2, \ldots, p_S)$, S denotes the total number of images
OUTPUT: Trained model
Initialize: Application Table II, $best_{acc} = 0$
for epoch = 0,1,2,...,20 do
  for iteration = 0,1,2,...,S/B do (Train)
    Extraction characteristics: Application Algorithm 1
    Classification: Expected distribution $(x_1, \ldots, x_n)$
    Loss = $L_(y_i, x_i)$, gradients = g_t
    parameters update: $W_{t+1} = W_t - \frac{\lambda \hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}$
  for iteration = 0,1,2,...,S/B do (Validation)
    Extraction characteristics: Application Algorithm 1
    Classification: Expected distribution $(x_1, \ldots, x_n)$
    Loss = $L_(y_i, x_i)$, no gradient calculation
  Calculation accuracy: acc = $f(y_i, x_i)$
  if $acc \geq best_{acc}$
    save model
RETURN: best model

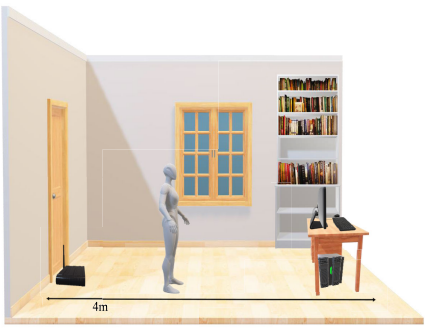


Fig. 6. Test environment.

1) Self-Test Dataset: Fig. 6 depicts a simplified laboratory layout. In the experiment, the router serves as a transmitter, while the computer serves as a receiver. The transmitter and receiver were 4 m apart. The CSI acquisition tool is installed on the computer, which is running Ubuntu 14.04. The receiver collects data at a sampling frequency of 100 Hz and with a total of 30 subcarriers.

Five volunteers and one experimenter participated in the experiment. The volunteers stood in the center of the transceiver and performed six actions, bending, drinking, nodding, squat, drawing the letter O, and drawing the letter B. Each volunteer performed the same action 20 times. About 100 samples of each action are collected. Each sample can be acquired with 30 subcarriers, each of which can be converted into an image, in fact 3000 samples per action.

2) Public Datasets: To assess ImgFi's overall performance, three public available high-quality datasets, WiAR [22], SAR [23], and Widar3.0 [28], are also used. WiAR collects WiFi signals reflected by 16 activities; SAR collects WiFi signals in response to six actions performed by nine volunteers over 6 days, and Widar3.0 collects six action signals from five volunteers at various locations and antenna orientations. Table III lists the specific action categories and dataset size.

TABLE III
DATASETS' DETAILS

Data Set	label	dataset size
Wiar [22]	two hands wave, high throw,horizontal arm wave, draw tick, toss paper,walk, side kick, bend, forward kick, drink water, sit down, draw-X, phone call, hand clap, high arm wave, squat	60,000 images
SAR [23]	clapping, falling, jumping, sitting, walking, waving	56,000 images
Widar [28]	push, sweep, clap, slide, draw-Z, draw-N	80,000 images
Self-test Dataset	bend, drink, nodd, squat, drawing-O, draw-B	18,000 images

B. Model Evaluation Details

The metrics used to evaluate the proposed framework are the standard accuracy, precision, recall, and F1 values. For the multiclassification problem, considering that the dataset samples are balanced, we first calculate the accuracy, precision, recall, and F1 values for each category and then calculate the arithmetic mean. The accuracy of each category can be expressed as follows:

$$\text{Acc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{29}$$

where TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative, respectively.

Each type of precision can be expressed as follows:

$$\text{Prec} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{30}$$

The recall rate per category is expressed as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{31}$$

Each type of F1 can be expressed as follows:

$$\text{F1} = \frac{2 \times \text{Prec} \times \text{Recall}}{\text{Prec} + \text{Recall}}. \tag{32}$$

Suppose the number of categories is N and the accuracy, precision, recall, and F1 values are expressed as follows:

$$\text{Macro_Acc} = \frac{1}{N} \sum_{i=1}^{N} \text{Acc} \tag{33}$$

$$\text{Macro_Prec} = \frac{1}{N} \sum_{i=1}^{N} \text{Prec} \tag{34}$$

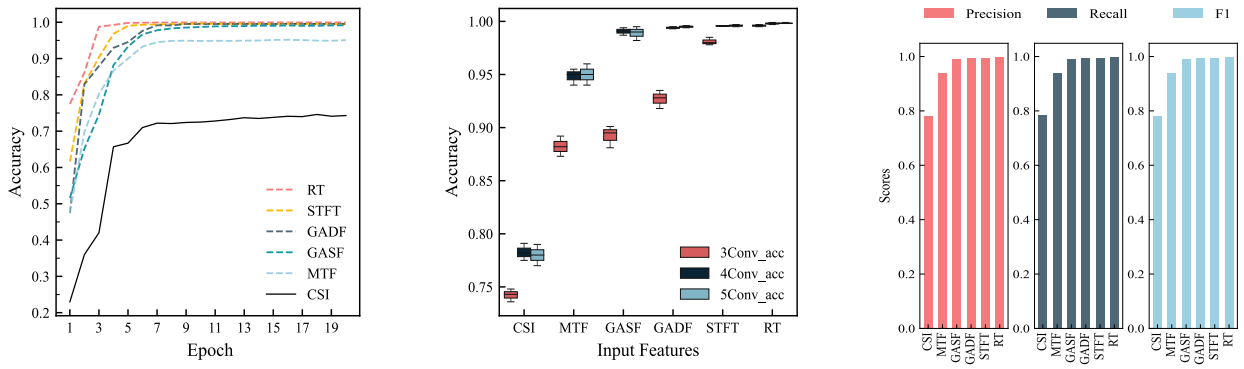
$$\text{Macro_Recall} = \frac{1}{N} \sum_{i=1}^{N} \text{Recall} \tag{35}$$

$$\text{Macro_F1} = \frac{1}{N} \sum_{i=1}^{N} \text{F1}. \tag{36}$$

Besides, we further evaluate the number of parameters and FLOPs of the model. The former is associated with the model size, and the latter is the computational complexity. FLOPs depend on the number of layers L and the computational complexity of each layer FLOPs. The FLOPs of the ReLU layer are much smaller than those of the convolutional and fully connected layers. Therefore, the computation of the ReLU layer is ignored in this article. The convolutional layer FLOPs denoted by F_c are calculated as follows:

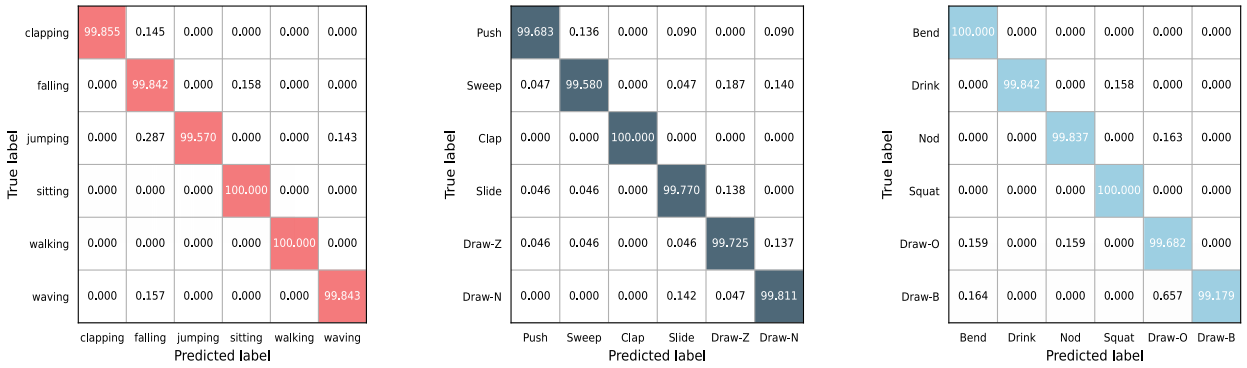
$$F_c = (2C_i \times K^2 - 1) \times H_o \times W_o \times C_o \tag{37}$$

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2026 at 12:51:29 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 7. (a) Training process with different input features. (b) Comparison of results with different input features and different numbers of convolution layers. (c) Comparison of different input characteristics.



(a) (b) (c)

Fig. 8. Confusion matrix for different datasets. (a) WiAR. (b) Widar3.0. (c) Self-test dataset.

where C_i and C_o are the input channel and output channel, respectively, k is the kernel size, and (H_o, W_o) is the size of the output feature map. The FLOPs of the linear layer are expressed by F_l and are computed as follows:

$$F_l = (2C_i - 1) \times C_o. \tag{38}$$

The BN layer's FLOPs F_b are expressed by

$$F_b = 2 * C_o * H_o * W_o. \tag{39}$$

The total FLOPs F_m is $F_c + F_l + F_b$, and the unit GFLOPs used in this article are defined as GFLOPs $= 10^9$ FLOPs.

VII. Evaluation Results

A. Overall Performance of ImgFi

As shown in Fig. 7(a), we give a plot of the training results for different features under four layers of convolution. After using the imaging method, the accuracy is improved by about 0.2 compared with the original CSI. As explained in Section IV 2-D images can provide both spatial and temporal information. The best performance among the five conversion methods is the RP-based one, which achieves a perfect accuracy of 0.998.

To further illustrate the potential of ImgFi, we did three experiments under three convolution layers, four convolutional layers, and five convolutional layers, and the results are shown in Fig. 7(b). Based on the RP method, the three-layer convolution can achieve an accuracy of more than 99.5%. With four-layer convolution, MTF can achieve 95% accuracy, and the remaining four imaging methods can achieve more than 99.0% accuracy. Compared with four-layer convolution, the five-layer convolution effect is not much improved.

Fig. 7(c) further shows the comparison of the results of the imaging-based method and the original CSI under four layers of convolution. We calculated the precision, recall, and F1 score for each feature and can see that ImgFi achieves consistently high performance. Specifically, RP outperforms STFT, GADF, and GASF, and all these four methods outperform MTF. As mentioned earlier, this is attributed to the difference in conversion methods. Fig. 8(a)–(c) further shows the RP-based confusion matrix under different datasets and the value of each element retains three decimal places. We provide average accuracy rates for all the gestures, and it can be seen that ImgFi achieves consistently high accuracy rates for all the gestures.

To further illustrate the potential of ImgFi, we compare the performance of the model under different training set samples. As shown in Fig. 9, using only 30% of the training samples, ImgFi converted by RT can achieve 92.0% recognition accuracy.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2026 at 12:51:29 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON WITH WELL-KNOWN DL MODELS

Datasets	Shufflenet			VGG19			ResNet18			ResNet50			ImgFi		
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
WiAR	99.4	99.4	99.4	99.8	99.7	99.7	99.8	99.8	99.7	99.8	99.8	99.8	99.9	99.8	**99.8**
SAR	99.4	99.4	99.3	99.7	99.8	99.7	99.8	99.8	99.8	99.8	99.8	99.8	99.8	99.8	**99.8**
Widar3.0	99.3	99.3	99.3	99.8	99.7	99.6	99.8	99.8	99.7	99.8	99.8	99.8	99.8	99.8	**99.8**
Self-test dataset	99.4	99.4	99.4	99.6	99.7	99.7	99.8	99.7	99.7	99.8	99.8	99.8	99.8	99.8	**99.8**

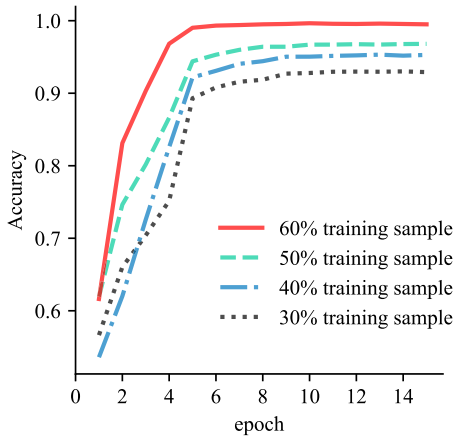


Fig. 9. Comparison results of different training sample proportions.

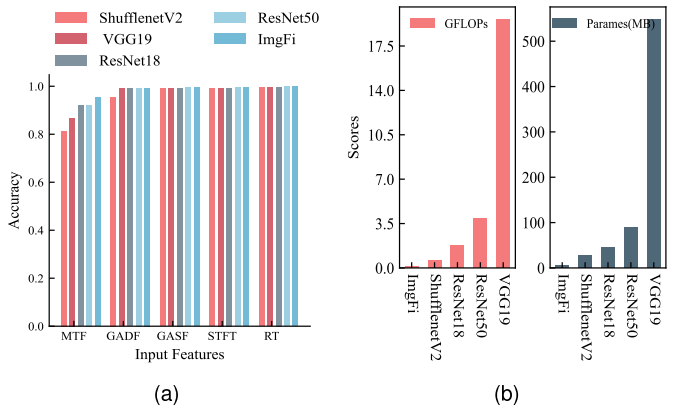


Fig. 10. (a) Comparison with classical DL models. (b) Comparison of model complexity.

B. Comparison With Classical DL Models

We further compared different DL models with the system performance shown in Table IV. Specifically, after imaging transformation, we can achieve high accuracy recognition rates using lightweight models. We input the RP images into four classical models, ResNet18, ResNet50, VGG19, and ShufflenetV2, respectively. We slightly outperformed shufflenet in the F1 score, which is comparable to ResNet and VGG19. This indicates that the RP images with different gestures are distinct enough that a more complex DL model does not affect.

Furthermore, we input STFT, GADF, GASF, MTF, and RP images into four classical models (ResNet50, ResNet18, VGG19, and ShufflenetV2), respectively. The results are shown in Fig. 10(a) in all the five images; ImgFi showed slightly better or comparable consistent results when comparing the classical models. This indicates that CSI imaging can express richer features and reduces the need for complex models.

The observation about the model resource consumption is shown in Fig. 10(b), where our model has only 6.5 MB size parameters and 0.1 GFLOPs size floating-point computation. Compared with the three classical DL models, our model has 90 times fewer parameters and requires 20 times fewer floating-point computations.

TABLE V
COMPARISON WITH EXISTING RESEARCH PROGRAMS

Approaches	Input Features	Architectures	Acc
Wi-Sensing	STFT	CNN+LSTM	96
MatNet-eCSI	CCF	CNN+LSTM	93
CSITime	CSI	InceptionTime	93
WiAReS	CSI	MLP+svm+RF	98.5
ImgFi	RT	CNN	**99.8**

C. Comparison With Existing Research Programs

As shown in Table V. Finally, we compare ImgFi with the most recent existing research solutions, including Wi-Sensing [27], CSITimecite [24], MatNet-eCSI [17], and WiAReS. Wi-Sensing, in particular, uses the STFT to convert temporal CSI into image data and achieves more than 96% classification accuracy by combining CNN and LSTM. MatNet-eCSI proposes a CSI correlation feature extraction (CCFE) method for improving and condensing activity-related information in the input signal, with a classification accuracy of 93%. CSITime achieves 98% recognition accuracy using a label blending strategy combined with an improved InceptionTime network architecture. WiAReS proposes an integrated architecture fusing multilayer perception (MLP), random forest (RF), and SVM to achieve 98.85% recognition accuracy. As a comparison, we surpassed the existing scheme with only four layers of convolution.

VIII. CONCLUSION

In this article, we propose a CSI-HAR framework: ImgFi whose main difference from the existing work is including a data conversion model. Specifically, before using the CNN for feature extraction, the original CSI is converted into the image. There are five conversion methods: recursive map, GASF, GADF, Markov variational field, and STFT, which are

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2026 at 12:51:29 UTC from IEEE Xplore. Restrictions apply.

used by ImgFi separately to achieve different performances. This transformation can preserve the time-series nature of the CSI, explain signal amplitude relationships for arbitrary timestamps, and allow spatial–temporal modeling of the CSI using the powerful image feature extraction capabilities of the 2D-CNN. The experimental results consistently show that ImgFi achieves high accuracy on all four datasets when a simple DL model is used. Specifically, based on the RP method, three-layer convolution can achieve more than 99.5% accuracy. With four-layer convolution, MTF can achieve 95% accuracy, and the remaining four imaging methods can achieve more than 99.0% accuracy. That is, the proposed ImgFi provides a standard and scalable solution framework for the CSI-HAR. In the future, we will design new models based on ImgFi to reduce the number of training samples and realize the cross-domain CSI-HAR.

REFERENCES

[1] Z. Sun, Q. Ke, H. Rahmani, M. Bennamoun, G. Wang, and J. Liu, "Human action recognition from various data modalities: A review," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3200–3225, Mar. 2023.

[2] Y. Wang, S. Cang, and H. Yu, "A survey on wearable sensor modality centred human activity recognition in health care," *Expert Syst. Appl.*, vol. 137, pp. 167–190, Dec. 2019.

[3] S. K. Yadav, K. Tiwari, H. M. Pandey, and S. A. Akbar, "A review of multimodal human activity recognition with special emphasis on classification, applications, challenges and future directions," *Knowl.-Based Syst.*, vol. 223, Jul. 2021, Art. no. 106970.

[4] A. Bulling, U. Blanke, and B. Schiele, "A tutorial on human activity recognition using body-worn inertial sensors," *ACM Comput. Surv.*, vol. 46, no. 3, pp. 1–33, Jan. 2014.

[5] G. Gkioxari, R. Girshick, P. Dollár, and K. He, "Detecting and recognizing human-object interactions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8359–8367.

[6] J. Han et al., "Twins: Device-free object tracking using passive tags," *IEEE/ACM Trans. Netw.*, vol. 24, no. 3, pp. 1605–1617, Jun. 2016.

[7] R. Du et al., "An overview on IEEE 802.11bf: WLAN sensing," 2022, *arXiv:2207.04859*.

[8] Y. Ma, G. Zhou, and S. Wang, "WiFi sensing with channel state information: A survey," *ACM Comput. Surv.*, vol. 52, no. 3, pp. 1–36, May 2020.

[9] S. Yousefi, H. Narui, S. Dayal, S. Ermon, and S. Valaee, "A survey on behavior recognition using WiFi channel state information," *IEEE Commun. Mag.*, vol. 55, no. 10, pp. 98–104, Oct. 2017.

[10] Y. Wang, J. Liu, Y. Chen, M. Gruteser, J. Yang, and H. Liu, "E-eyes: Device-free location-oriented activity identification using fine-grained WiFi signatures," in *Proc. 20th Annu. Int. Conf. Mobile Comput. Netw.*, Sep. 2014, pp. 617–628.

[11] W. Wang, A. X. Liu, M. Shahzad, K. Ling, and S. Lu, "Device-free human activity recognition using commercial WiFi devices," *IEEE J. Sel. Areas Commun.*, vol. 35, no. 5, pp. 1118–1131, May 2017.

[12] G. Lin et al., "Human activity recognition using smartphones with WiFi signals," *IEEE Trans. Human-Mach. Syst.*, vol. 53, no. 1, pp. 142–153, Feb. 2023.

[13] M. A. A. Al-Qaness, "Device-free human micro-activity recognition method using WiFi signals," *Geo-Spatial Inf. Sci.*, vol. 22, no. 2, pp. 128–137, Apr. 2019.

[14] S. Arshad et al., "Wi-chase: A WiFi based human activity recognition system for sensorless environments," in *Proc. IEEE 18th Int. Symp. A World Wireless, Mobile Multimedia Netw. (WoWMoM)*, Jun. 2017, pp. 1–6.

[15] Z. Hao, Y. Duan, X. Dang, Y. Liu, and D. Zhang, "Wi-SL: Contactless fine-grained gesture recognition uses channel state information," *Sensors*, vol. 20, no. 14, p. 4025, Jul. 2020.

[16] B. A. Alsaify, M. M. Almazari, R. Alazrai, S. Alouneh, and M. I. Daoud, "A CSI-based multi-environment human activity recognition framework," *Appl. Sci.*, vol. 12, no. 2, p. 930, Jan. 2022.

[17] Z. Shi, J. A. Zhang, R. Y. Xu, and Q. Cheng, "Environment-robust device-free human activity recognition with channel-state-information enhancement and one-shot learning," *IEEE Trans. Mobile Comput.*, vol. 21, no. 2, pp. 540–554, Feb. 2022.

[18] P. Fard Moshiri, R. Shahbazian, M. Nabati, and S. A. Ghorashi, "A CSI-based human activity recognition using deep learning," *Sensors*, vol. 21, no. 21, p. 7225, Oct. 2021.

[19] S. Arshad, C. Feng, R. Yu, and Y. Liu, "Leveraging transfer learning in multiple human activity recognition using WiFi signal," in *Proc. IEEE 20th Int. Symp. World Wireless, Mobile Multimedia Netw. (WoWMoM)*, Jun. 2019, pp. 1–10.

[20] B. Li, W. Cui, W. Wang, L. Zhang, Z. Chen, and M. Wu, "Two-stream convolution augmented transformer for human activity recognition," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, no. 1, 2021, pp. 286–293.

[21] W. Cui, B. Li, L. Zhang, and Z. Chen, "Device-free single-user activity recognition using diversified deep ensemble learning," *Appl. Soft Comput.*, vol. 102, Apr. 2021, Art. no. 107066.

[22] L. Guo et al., "WiAR: A public dataset for WiFi-based activity recognition," *IEEE Access*, vol. 7, pp. 154935–154945, 2019.

[23] J. K. Brinke and N. Meratnia, "Scaling activity recognition using channel state information through convolutional neural networks and transfer learning," in *Proc. 1st Int. Workshop Challenges Artif. Intell. Mach. Learn. Internet Things*, Nov. 2019, pp. 56–62.

[24] S. K. Yadav et al., "CSITime: Privacy-preserving human activity recognition using WiFi channel state information," *Neural Netw.*, vol. 146, pp. 11–21, Feb. 2022.

[25] J. Schäfer, B. R. Barrsiwal, M. Kokhkharova, H. Adil, and J. Liebehenschel, "Human activity recognition using CSI information with Nexmon," *Appl. Sci.*, vol. 11, no. 19, p. 8860, Sep. 2021.

[26] Z. Shi, J. A. Zhang, R. Xu, and Q. Cheng, "Deep learning networks for human activity recognition with CSI correlation feature extraction," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2019, pp. 1–6.

[27] H. Lee, C. R. Ahn, and N. Choi, "Fine-grained occupant activity monitoring with Wi-Fi channel state information: Practical implementation of multiple receiver settings," *Adv. Eng. Informat.*, vol. 46, Oct. 2020, Art. no. 101147.

[28] Y. Zhang et al., "Widar3.0: Zero-effort cross-domain gesture recognition with Wi-Fi," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 8671–8688, Nov. 2022.

[29] E. Shalaby, N. ElShennawy, and A. Sarhan, "Utilizing deep learning models in CSI-based human activity recognition," *Neural Comput. Appl.*, vol. 34, no. 8, pp. 5993–6010, Apr. 2022.

[30] J. Yang, H. Zou, Y. Zhou, and L. Xie, "Learning gestures from WiFi: A Siamese recurrent convolutional architecture," *IEEE Internet Things J.*, vol. 6, no. 6, pp. 10763–10772, Dec. 2019.

[31] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[32] Z.-Q. Zhao, P. Zheng, S.-T. Xu, and X. Wu, "Object detection with deep learning: A review," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 11, pp. 3212–3232, Nov. 2019.

[33] A. Sharma, E. Vans, D. Shigemizu, K. A. Boroevich, and T. Tsunoda, "DeepInsight: A methodology to transform a non-image data to an image for convolution neural network architecture," *Sci. Rep.*, vol. 9, no. 1, pp. 1–7, Aug. 2019.

[34] Z. Wang and T. Oates, "Imaging time-series to improve classification and imputation," in *Proc. 24th Int. Conf. Artif. Intell. (IJCAI)*, 2015, pp. 3939–3945.

[35] B. Bertalanic, M. Meza, and C. Fortuna, "Resource-aware time series imaging classification for wireless link layer anomalies," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Feb. 17, 2022, doi: 10.1109/TNNLS.2022.3149091.

[36] H. Xu et al., "Human activity recognition based on Gramian angular field and deep convolutional neural network," *IEEE Access*, vol. 8, pp. 199393–199405, 2020.

[37] J. Eckmann et al., "Recurrence plots of dynamical systems," *World Sci. Ser. Nonlinear Sci. Ser. A*, vol. 16, pp. 441–446, Sep. 1995.

[38] Z. Ahmad and N. Khan, "Inertial sensor data to image encoding for human action recognition," *IEEE Sensors J.*, vol. 21, no. 9, pp. 10978–10988, May 2021.

[39] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on June 26,2026 at 12:51:29 UTC from IEEE Xplore. Restrictions apply.